MIL-G-45851D
11 July 1985
SUPERSEDING
MIL-G-45851C
14 July 1978

MILITARY SPECIFICATION

GRINDING MACHINE, CYLINDER HEAD (WET TYPE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a stationary, motor driven, surface grinding machine for wet grinding cylinder heads, manifolds, and engine blocks for in-line and V-type internal combustion engines and similar flat surfaces.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-1807	- Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).

MILITARY

MIL-T-152	- Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-M-18058	- Machinery, Metal and Woodworking, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-889 - Dissimilar Metals.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.
AGMA 390.03 - Gear Handbook, Volume 1, Gear Classification, Materials, and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, 1901 N. Fort Myer Drive, Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B7.1-78 - Safety Requirements for the Use, Care, and Protection of Abrasive Wheels.
ANSI B11.9-75 - Safety Requirements for the Construction, Care, and Use of Grinding Machines.
ANSI B74.2-74 - Specifications for Grinding Wheels, Shapes and Sizes, Mounted Wheels, Sizes and Identification.
ANSI Y14.5M-82 - Dimensioning and Tolerancing.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.
ASTM E380-84 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches.
ISO 1122/1 - Glossary of Gear Terms - Part 1: Geometrical Definitions.

(Application for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one or more grinding machines for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The grinding machine shall be new and of the manufacturer's latest design meeting the performance requirements specified herein. Components of the machine shall be designed to be of the size and strength necessary to sustain the maximum allowable operating load with maximum efficiency and minimum wear. The grinding machine shall be designed with a grinding wheel spindle mounted in the vertical or horizontal position and located either above or below the parts being ground. The grinding wheel head, or table, shall provide the traverse and feed motions required. The machine shall be provided with adjustable stops that reverse travel automatically.

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3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the grinding machine. In this specification, all measurements, dimensions, sizes and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-84.

3.2.2 Measuring and indicating devices. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI systems of measurement. When a dual US and SI system is furnished, dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.3 Maintainability. Design of the grinding machine and the placement of adjusting features shall insure that operational maintenance capability shall be reduced to the simplest form of procedures, requiring a minimum of skills and common tools. Parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair without modification, distortion, or damage to the machine or any machine part or component.

3.2.4 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible without requiring disassembly of functional parts. Lubrication reservoirs, if applicable, shall have means for checking levels.

3.2.5 Gears. All gears and pinion designs selected for use in the grinding machine or its components shall be manufactured to either English (USA) standard inch diametral pitch per AGMA 360.02 and AGMA 390.03 or Metric (SI) module ratio per the International Organization for Standardization (ISO) ISO 54 and ISO 1122/1 and ASTM E380-84.

3.2.6 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.7 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.8 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards. Guards shall be removable to provide easy access to guarded parts. The grinding machine shall comply with the general safety and health requirements promulgated under OSHA 2206 that are applicable to the machine itself. In addition, the grinding machine shall comply with those requirements of ANSI B11.9 that are specified therein as the responsibility of the machine manufacturer. In the event of conflict between the OSHA and ANSI

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standards, the OSHA requirements shall apply. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the machine tool industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. The grinding machine shall be constructed to be capable of performing the operations specified herein. The machine shall be complete so that, when installed and connected to a proper power source, the machine may be used for all operations for which it was designed.

3.4.1 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.4.2 Welding, brazing, or soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. The methods for preventing loss of tightness or adjustment shall be in accordance with acccepted engineering standards and practice. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.4.4 Surfaces. All castings, forgings, molded parts, stampings, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.4.5 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in all soldering operations.

3.5 Performance and product characteristics. The grinding machine shall be capable of wet grinding aluminum, alloy steel, or cast iron cylinder heads,

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manifolds, and engine blocks of in-line and V-type internal combustion engines and other flat-surfaced parts.

3.5.1 Accuracy. The grinding machine shall be capable of grinding a surface flatness which shall not deviate more than 0.002 inch from the plane of the surface being ground. The surfaces shall parallel the center line of main bearings within 0.002 inch to assure perpendicular alignment of assembled components when measured as defined in ANSI Y14.5M-82.

3.5.2 Capacity. The grinding machine shall have a minimum work capacity not less than 46 inches in length by 14 inches wide. The vertical height capacity, in inches, shall be as specified by the procuring activity (see 6.2.1).

3.6 Components. The grinding machine shall consist of, but may not necessarily be limited to, the principal components covered in 3.6.1 through 3.6.6.1.

3.6.1 Base. The grinding machine base shall be constructed of metal and shall be sufficiently rigid to meet the performance characteristics specified in 3.5. Leveling devices and hold-down bolt holes shall be provided.

3.6.2 Working area. The working area shall be of sufficient capacity to accommodate the minimum work size specified in 3.5.2 and shall provide support for the work. Necessary provisions shall be furnished for holding the work securely in place for the grinding operation.

3.6.3 Grinding wheel. Grinding wheels for vertical grinding shall be either hollow center cupped or segmented types, not less than 14 inches in diameter. The wheels shall comply with ANSI B7.1. Grinding wheels for horizontal grinding shall be either straight (Type 1) or recessed (two sides, Type 7) complying with ANSI B74.2. If separate grinding wheels are necessary for grinding various metals (see 3.5), one wheel of each such type shall be provided with each machine.

3.6.4 Wheel spindle. The wheel spindle shall rotate in bearings sealed against abrasive dust. Vertical feed shall be accomplished by micrometer type adjustment graduated in 0.001 inch increments. The feed adjustment shall be accessible to the operator.

3.6.5 Coolant system. A gear or impeller type pump shall be provided to force the coolant to the grinding wheel, to flush the wheel free of grinding residue, and to cool the part being ground. The capacity of the coolant system shall be of sufficient size to provide an adequate amount of coolant during continuous operation of the machine. Splash guards or splash rings shall direct the coolant back to the reservoir and prevent coolant loss. Provision shall be made for the coolant to settle the residue or to be filtered before its return to the coolant reservoir. Means shall be provided for complete flushing and draining of the coolant reservoir.

3.6.6 Electrical equipment. Electrical equipment shall conform to NFPA 79.

3.6.6.1 Motors. The main drive motor shall be rated at not less than 5 1/2 horsepower, continuous duty, conforming to CC-M-1807, with class B insulation.

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The main motor shall be wired to operate from 230-460 volts, three phase, 60 hertz power source, and initially wired to operate from 230 volts. Feed and pump motors shall be induction, squirrel cage type, conforming to CC-M-1807.

3.6.6.2 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.7 Standard equipment. The grinding machine shall be furnished with all equipment, work holding provisions, and attachments necessary to comply with the performance requirements specified in 3.5 through 3.5.2. Standard equipment shall include but shall not be limited to dresser, wheel, guard, wheel dresser, clamps, blocks, holding fixtures, and jaw fingers.

3.8 Spare parts. Unless otherwise specified (see 6.2.1), the following spare parts shall be furnished with each grinding machine:

(a) One extra belt of each type used.
(b) One extra grinding wheel of each type required (see 3.6.3).

3.9 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.10 Product identification. The grinding machine shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.11 Workmanship. Standards of workmanship shall assure that the grinding machine shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Accuracy test (see 4.6.1).
(c) Packaging inspection (see 4.7).

4.4.1 Sampling. When required (see 6.2.1), sampling for quality conformance inspection shall be as specified in MIL-STD-105 using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Product examination (4.5)	II	1.0
Accuracy test (4.6.1)	II	1.0
Packaging inspection (4.7)	S-4	4.0

4.5 Product Examination. Visually and manually examine the grinding machine to determine conformance with the requirements of 3.2 through 3.3, 3.4 through 3.4.5, and 3.6 through 3.11. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the grinding machine passed the examinations prescribed by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Accuracy test. The grinding machine shall be set up and operated in accordance with the manufacturer's instructions. A cylinder head from a standard size, commercial, six cylinder engine shall be ground to the extent necessary to refinish the entire surface. The ground surface shall be tested with precision equipment to determine conformance to the requirements (see 3.5.1).

4.6.2 Performance test. The grinding machine shall be operated in accordance with the manufacturer's instructions while using the standard equipment noted in 3.7. The wheel of the grinder shall first be dressed and the grinding machine shall then be used to completely refinish the gasket bearing surfaces of a six cylinder engine block, a V-8 engine block, the heads from those two engine blocks, and the intake and exhaust manifolds from both those engines.

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The grinding wheel shall be redressed as required. The refinished areas shall then be tested as specified in 4.6.1.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation and for Levels A and B packing and marking shall be in accordance with MIL-M-18058.

5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. The grinding machine covered by this specification is intended to be used primarily to recondition the gasket bearing surfaces of cylinder heads, engine blocks, manifolds, and other similar flat surfaces.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Measuring and indicating device graduation(s) required (see 3.2.2).
(d) Exceptions and additional safety/health requirements when required (see 3.2.8 and 6.4).
(e) Vertical height capacity, in inches, required (see 3.5.2).
(f) Spare parts if different (see 3.8).
(g) Painting and finishing if different (see 3.9).
(h) Product identification if different (see 3.10).
(i) First article inspection if different (see 4.3).
(j) Quality conformance inspection if different (see 4.4).
(k) Sampling when required (see 4.4.1).
(l) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract (see 6.3.1).

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

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6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - AL
Navy - YD
Air Force - 99

Review activities:

Navy - MC
Air Force - 82

Preparing activity:

Army - AL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-45851D	Grinding Machine, Cylinder Head (Wet Type)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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